

Final Year Project Report

AI Powered Multi-Prognosis Disease Prediction Image Classification

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Abstract

This project presents the design and development of a multi-prognosis disease prediction system using a Random Forest Classifier, trained on binary symptom-based data. The system allows users to input 4–5 symptoms and receive a predicted disease along with detailed information on precautions, medications, and disease descriptions. Unlike black-box deep learning models, Random Forest provides interpretability and high performance on structured, tabular medical data. The user interface, built using Streamlit, ensures an accessible experience for real-time prediction. Further enhancing functionality, the system integrates a medical AI chatbot powered by LangChain and Groq's implementation of the LLaMA model. The chatbot can respond to user queries using a custom knowledge base built from PDF medical documents via FAISS-based semantic retrieval. The model achieved a testing accuracy of 96.7%, with strong F1 scores across multiple disease classes. Evaluations of the chatbot also showed high response accuracy and relevance. This hybrid diagnostic-support tool demonstrates the feasibility of building lightweight, interpretable, and interactive healthcare AI systems suitable for both clinical support and patient education in resource-constrained settings.

Background Research and Domain Analysis

1.1 Introduction to AI in Healthcare

The adoption of artificial intelligence (AI) in healthcare has grown significantly in the past decade. Driven by improvements in computational power and the availability of large health datasets, AI systems are now able to assist clinicians with diagnosis, prognosis, personalized treatment planning, and risk stratification. One of the core domains within AI that has demonstrated substantial success is machine learning (ML) particularly in tasks that require classification of health conditions based on observable indicators like symptoms or lab results.

Within this domain, the use of symptom-based disease prediction models has become a focal point of research and deployment. These models aim to mimic the diagnostic process of a clinician by analyzing symptoms to predict the most likely disease. However, doing this in an accurate, scalable, and interpretable manner presents unique challenges, which this project seeks to address.

1.2 Problem Space: Diagnostic Bottlenecks in Healthcare

Healthcare systems across both developed and developing nations face significant diagnostic bottlenecks that impede timely and accurate disease identification. These bottlenecks arise from a confluence of systemic challenges, including workforce shortages, geographic limitations, and varying levels of clinician expertise. As a result, the diagnostic process is often delayed, particularly for early-stage diseases that may present with non-specific or overlapping symptoms. This delay can lead to suboptimal patient outcomes, especially in cases where prompt intervention is critical.

Challenges in Diagnosis: Overlapping Symptoms and Workforce Shortages

One of the fundamental challenges in medical diagnosis is the frequent overlap of symptoms among various diseases. Symptoms such as fever, headache, fatigue, and nausea are common across a wide range of conditions, from viral infections like influenza and COVID-19 to bacterial infections like typhoid or malaria. Early-stage diseases often exhibit subtle or vague symptoms that are not distinct enough to warrant immediate and clear diagnosis. As such, clinicians face the daunting task of distinguishing between these conditions, requiring extensive diagnostic testing, which may not always be readily available or feasible due to resource limitations.

The situation is exacerbated in rural areas and remote regions, where there may be a lack of specialized medical professionals or diagnostic equipment. In many developing countries, healthcare facilities are poorly equipped to conduct the advanced tests required for accurate diagnosis, and patients often need to travel long distances to access healthcare services. According to the World Health Organization (WHO), approximately half of the world's population lacks access to essential health services, with a disproportionate impact on rural and underserved communities (WHO, 2021). This geographic disparity results in delayed diagnoses and can increase the risk of disease progression, leading to worse health outcomes.

Similarly, even in developed countries, the variability in clinician experience and the strain on healthcare systems can lead to diagnostic delays. The COVID-19 pandemic highlighted the vulnerabilities in healthcare infrastructure, as hospitals and clinics were overwhelmed by an influx of patients, many of whom presented with similar flu-like symptoms. Diagnostic tools, such as polymerase chain reaction (PCR) tests, were overburdened, and the rapid need for diagnosis often outstripped the capacity of medical professionals and diagnostic laboratories. This situation underscored the need for more efficient diagnostic systems capable of triaging and identifying likely diseases quickly and accurately, even in resource-constrained environments.

The Promise of AI in Addressing Diagnostic Bottlenecks

In this context, artificial intelligence (AI) systems present a promising solution to alleviate diagnostic bottlenecks. AI-based systems can process large amounts of data rapidly and provide diagnostic suggestions based on limited user input. By leveraging machine learning (ML) algorithms that can analyze symptoms, medical history, and other relevant factors, AI systems have the potential to assist healthcare workers in making accurate diagnoses, even in settings where resources are limited.

AI-based diagnostic tools can be particularly valuable in triaging patients, helping prioritize cases that require immediate attention and directing patients to appropriate care pathways. Such systems can be deployed in primary care settings, mobile health applications, or rural clinics where trained healthcare providers may not be readily available. By reducing the reliance on expensive and time-consuming diagnostic tests, AI models can help speed up the diagnostic process and provide early intervention for diseases that might otherwise be overlooked.

However, for AI-based systems to be effective in real-world healthcare settings, they must meet several key requirements:

1. **Interpretability:** AI models, particularly in healthcare, must be interpretable. Clinicians need to understand how the model arrived at its diagnosis in order to trust its predictions and incorporate them into clinical decision-making. Unlike black-box models that provide little insight into the decision-making process, models like Random Forest offer feature importance scores, which allow healthcare providers to understand which symptoms were most influential in making a prediction. This transparency is essential for ethical AI adoption in healthcare, where decisions based on model predictions can directly impact patient outcomes.
2. **Lightweight and Efficient:** The AI system must be lightweight and computationally efficient. This is especially important in low-resource settings, where devices with limited processing power or no access to high-performance computing infrastructure are common. The system should be able to function effectively on basic hardware, such as mobile phones or low-cost diagnostic devices, without relying on expensive cloud-based infrastructure. Ensuring the system's efficiency will also make it more scalable, allowing for widespread deployment in areas that need it most.
3. **Supportive Information:** In addition to providing diagnostic predictions, the AI system should offer meaningful support information. For example, once a probable disease is identified, the system should provide recommendations regarding medications, precautions, and preventive measures. This could help both patients and healthcare workers take appropriate steps while awaiting confirmation through further testing or while managing minor health issues. Information about common treatments and lifestyle changes could be integrated into the AI system's response, empowering users with actionable knowledge and enhancing patient education.

Real-World Applications of AI in Diagnosing Overlapping Symptoms

Several real-world applications demonstrate how AI can help mitigate diagnostic bottlenecks. For instance, symptom-checker applications, such as those developed by Ada Health and Babylon Health, have gained significant attention in recent years. These applications use AI algorithms to assess user-reported symptoms and suggest potential diagnoses. They have been particularly valuable during the COVID-19 pandemic, where millions of people worldwide sought information and guidance about their symptoms through AI-powered systems.

These applications use a similar approach to the one outlined in this project: processing a small set of symptoms to generate a list of possible conditions. However, they are generally constrained to specific symptom sets or geographic regions. The advantage of using an AI system that accepts a broader range of symptoms and provides more nuanced diagnostic information—such as the likelihood of each condition—has the potential to offer even greater support to healthcare providers in diverse settings.

The integration of conversational AI further enhances the diagnostic tool's value by allowing users to interact with the system in a more human-like manner. For example, a user could ask follow-up questions about their symptoms, medications, or preventive measures. The AI-powered chatbot could then provide context-specific answers based on a curated knowledge base. This makes the tool not just a diagnostic aid but a more holistic healthcare companion.

Addressing the Challenges: Scalability and Ethical Considerations

While the promise of AI in diagnosing diseases based on symptoms is vast, it is also essential to address some key challenges. Ethical considerations, particularly regarding patient data privacy and the risk of misdiagnosis, are paramount. AI models must be designed to ensure that they do not replace human judgment but instead act as support tools to enhance decision-making. Disclaimers should be clearly communicated to users, emphasizing that AI outputs are suggestions and not definitive diagnoses.

Moreover, AI systems must be continuously updated to account for emerging diseases, new research findings, and evolving medical guidelines. This requires ongoing data collection, collaboration with medical professionals, and validation of the system's outputs. As the system evolves, it should also incorporate feedback loops to improve its predictions and responses based on real-world data.

Conclusion: The Future of AI in Healthcare Diagnostics

AI has the potential to revolutionize diagnostic practices, particularly in settings where resources are scarce or diagnostic capabilities are limited. By processing a small number of symptoms, AI models can offer fast, accurate predictions and provide valuable support information to both healthcare workers and patients. To ensure their success, these systems must be interpretable, lightweight, and capable of delivering meaningful diagnostic and treatment-related insights. As the technology continues to advance, AI will play an increasingly pivotal role in bridging the diagnostic gaps in healthcare systems worldwide.

1.3 Literature Review of Symptom-Based ML Systems

Overview of Machine Learning in Symptom-Based Medical Diagnosis

Machine learning (ML) has become a cornerstone of modern medical diagnosis, particularly in the context of symptom-based disease prediction. Accurate and timely disease diagnosis is often hindered by overlapping symptoms and ambiguous patient reports. Therefore, ML systems that can analyze a limited set of symptoms and predict the most likely diseases have significant potential in reducing diagnostic delays and improving patient outcomes. These systems are valuable in scenarios where direct access to specialists is limited, and even in resource-constrained environments like rural areas or low-resource settings, ML models can serve as a critical first step in diagnosis.

Among the various machine learning techniques, ensemble methods—specifically Random Forest (RF)—have proven to be particularly effective in this domain. Ensemble methods combine the predictions of multiple individual models to produce a stronger, more robust prediction. Random Forest, an ensemble of decision trees, is a go-to algorithm due to its robustness, accuracy, and interpretability. It excels at handling both binary and categorical data, which is prevalent in medical symptom classification. This section reviews key studies that demonstrate the application of Random Forest in predicting diseases based on symptoms, illustrating the method's strength in handling real-world data limitations, including noise, class imbalance, and feature redundancy.

Key Studies on Random Forest in Symptom-Based Diagnosis

Kibria and Matin (2022) explored the use of a fusion machine learning model that combined Random Forest with other algorithms to predict cardiovascular diseases (CVD). Their research demonstrated that Random Forest provided high accuracy and robustness, especially in multi-class classification tasks where diseases share similar symptoms. One of the key advantages noted was the model's performance compared to traditional methods such as logistic regression and Naive Bayes. Random Forest achieved superior results in distinguishing between multiple classes of CVDs based on limited input data (*Kibria and Matin, 2022*).

The fusion approach involved combining Random Forest with other ML algorithms, optimizing the system for both predictive accuracy and computational efficiency. This fusion enhanced the model's ability to handle noisy and incomplete data, a critical feature in healthcare settings where patient information can often be sparse or imprecise. The study's findings underscore Random Forest's versatility and reliability in diagnosing complex diseases like CVD, which is characterized by a broad spectrum of symptoms that overlap with other conditions. Moreover, the ability to manage various data types—such as patient demographics, medical history, and symptom reports—further validates Random Forest's robustness in real-world healthcare applications.

Chunduru et al. (2024) utilized Random Forest to predict chronic diseases based on a small set of user-reported symptoms. Their study highlighted the feasibility of using machine learning in low-resource environments, where patients might only remember or report a limited number of

symptoms. Despite the reduced input features, the Random Forest model demonstrated resilience, effectively handling the ambiguity and uncertainty inherent in real-world medical data (Chunduru et al., 2024).

The system was deployed using Flask, a lightweight web framework, which facilitated easy integration into mobile applications or rural clinics. The use of Random Forest in this context proved especially valuable, as it allowed for real-time predictions on devices with limited computational resources. The study showed that even when only a few symptoms were reported by patients, the model could still accurately predict chronic diseases such as diabetes or hypertension, which often manifest in early stages with mild or generalized symptoms. This work contributed to the growing body of evidence that Random Forest can operate effectively with minimal symptom input, making it an ideal choice for healthcare applications where data availability is restricted.

Ajabani (2023) proposed a supervised multi-class classifier using Random Forest to predict diseases from 133 binary symptom features. This study emphasized two important aspects: model explainability and user interface (UI) accessibility. These aspects are vital for deploying machine learning models in clinical settings where interpretability is essential for gaining clinician trust and regulatory approval. Random Forest's feature importance functionality allowed Ajabani to provide insights into which symptoms contributed most significantly to the model's predictions, thus improving the transparency of the diagnostic process (Ajabani, 2023).

Moreover, the study emphasized the development of a user-friendly interface that could be easily understood by non-experts. This is particularly important in healthcare applications, where models are often used by healthcare workers who may not have a deep technical understanding of the underlying algorithms. By combining a multi-class Random Forest model with an accessible UI, Ajabani's work further reinforced the need for explainability in machine learning systems, particularly when used in high-stakes domains like healthcare.

Meriç and Özer (2024) conducted a comparative study between multiple machine learning models, including Random Forest, logistic regression, support vector machines (SVM), and k-nearest neighbors (KNN), to evaluate their effectiveness in symptom-based disease prediction. The authors concluded that Random Forest offered a favorable balance between speed, accuracy, and interpretability, making it an ideal choice for medical diagnosis systems (Meriç and Özer, 2024).

The study also highlighted the challenges of working with symptom-based data, including class imbalance and symptom redundancy. These challenges make it difficult for many models to perform well, especially when diseases share a large number of overlapping symptoms. However, Random Forest's ability to handle noisy data, along with its robustness against overfitting, allowed

it to outperform other models in multi-class classification tasks. The study emphasized the importance of choosing the right model based on the problem context, particularly in healthcare, where decision-making is highly sensitive and must be both fast and accurate.

Patel et al. (2024) explored a hybrid approach to disease prediction, combining Random Forest with deep learning models. This combination aimed to leverage the strength of deep learning for capturing complex patterns and the interpretability of Random Forest for model transparency. The hybrid system performed well in predicting diseases from symptom data and was particularly effective in managing the trade-off between accuracy and *explainability* (Patel et al., 2024).

The researchers noted that while deep learning models like neural networks can capture more intricate relationships between symptoms, they often suffer from a “black-box” problem, making it difficult for clinicians to understand why a particular prediction was made. By combining these models with Random Forest, Patel et al. were able to provide high accuracy without sacrificing interpretability. This hybrid approach is particularly promising for systems that require both robust performance and transparency, as it offers the best of both worlds.

Random Forest for Symptom-Based Diagnosis: Advantages and Limitations

Random Forest is a powerful tool in symptom-based disease prediction due to its robustness, interpretability, and versatility. Its ability to handle both binary and categorical data makes it particularly suitable for medical diagnosis tasks, where symptoms are typically encoded as binary features (e.g., present or absent). Moreover, the model’s resistance to overfitting and its feature importance mechanism contribute to its appeal in healthcare settings where model transparency and interpretability are crucial for clinical adoption.

However, despite these advantages, Random Forest models also come with certain limitations. One of the primary challenges is the model’s reliance on large amounts of data to achieve high performance. In healthcare applications, acquiring sufficient and diverse data can be challenging, especially when dealing with rare diseases or incomplete medical records. Additionally, while Random Forest provides a high level of interpretability compared to deep learning models, it may still be difficult for non-experts to fully understand the intricacies of the model’s decision-making process.

Another challenge is the potential difficulty in handling highly dimensional data, especially when many features are irrelevant or redundant. Although Random Forest is robust to irrelevant features, there is still a need for efficient feature selection to reduce computational complexity and improve model performance.

Future Directions in Symptom-Based ML Systems

The studies reviewed in this section highlight the continued evolution of machine learning in healthcare, particularly in the context of symptom-based disease prediction. Future research may focus on enhancing the adaptability of Random Forest models to handle more complex, dynamic input features, such as symptom severity, duration, and patient demographics. Moreover, incorporating multi-modal data, such as patient medical history, lab results, and imaging data, could further improve the accuracy of symptom-based disease predictions.

In addition, the integration of explainable AI (XAI) techniques could help further bridge the gap between technical performance and clinical interpretability. As machine learning models become more sophisticated, ensuring that they remain interpretable and actionable will be critical to their adoption in clinical practice.

1.4 The Role of Multi-Class Classification

In real-world medical diagnosis tasks, the challenge often extends beyond distinguishing between just two outcomes. Instead, multiple potential diagnoses must be considered, making these problems inherently multi-class classification tasks. In healthcare, diseases often share overlapping symptoms, which complicates the diagnosis process and necessitates the use of models capable of handling multiple classes simultaneously. Traditional binary classification approaches are insufficient for this purpose, as they are designed to differentiate only between two distinct classes. When dealing with numerous potential diseases, a more robust approach is required.

Random Forest natively supports multi-class classification, making it an ideal choice for this task. Each decision tree in the Random Forest is trained to classify samples across all available disease classes. This approach contrasts with models that require a one-vs-all decomposition, which would train separate binary classifiers for each disease, thereby increasing both the training time and the complexity of the model. The ability to classify multiple classes directly within each tree allows Random Forest to perform multi-class classification more efficiently, improving both training time and inference performance. This efficiency is especially important when deploying the model in real-time systems, such as mobile health applications or telemedicine platforms, where speed and responsiveness are critical (Beniwal et al., 2020).

Moreover, Random Forest models are highly resistant to overfitting, especially compared to individual decision trees. In the medical context, where the input data—such as symptom reports—can be noisy or partially incorrect, this robustness is crucial. For instance, patients may report symptoms in ways that are ambiguous or inconsistent, making it challenging for simpler models to generalize effectively. Random Forest's ensemble approach, which combines multiple decision trees, helps mitigate overfitting by averaging out errors and reducing the impact of individual misclassifications. This characteristic makes Random Forest an ideal choice for medical diagnosis, where accuracy and resilience to data quality issues are paramount.

By directly supporting multi-class classification and offering resilience to noisy data, Random Forest is a powerful tool in medical AI applications, ensuring accurate predictions and quick decision-making, even in challenging real-world scenarios.

1.5 Integration with Conversational AI

One of the distinguishing features of this system is its integration of a medical chatbot, powered by LangChain and Groq's LLaMA model. This feature bridges the gap between machine learning predictions and user interaction.

Most ML-based diagnosis tools stop at prediction. However, users often have follow-up questions like:

- “How do I manage these symptoms?”
- “What medications are commonly used?”
- “Is this disease contagious?”

To answer these, the chatbot uses semantic search over a curated PDF knowledge base (e.g., WHO reports, medical textbooks), which is embedded into a FAISS vector store and queried using LangChain's retrieval-augmented generation (RAG) pattern.

This hybrid approach—combining structured ML with unstructured NLP—adds enormous value to the end-user experience.

1.5 Market Demand and Use Cases

The demand for telemedicine and self-diagnosis tools has surged, particularly in the wake of the COVID-19 pandemic, which highlighted the need for scalable, remote healthcare solutions. During the pandemic, healthcare systems across the globe were overwhelmed, pushing for more efficient ways to diagnose and manage patients remotely. This led to an increase in the adoption of AI-supported diagnostic tools, including symptom checkers and telemedicine platforms. Google's symptom-checker API and the rise of companies like Ada Health and Babylon Health demonstrate the commercial and clinical interest in AI-based diagnostic support tools that help patients understand their symptoms and suggest possible conditions.

As telemedicine and self-diagnosis tools become more mainstream, the market has evolved to demand lightweight, easily deployable solutions that can function effectively in environments with limited infrastructure. This project aligns well with these growing market needs by focusing on lightweight, offline-capable solutions. Such solutions are especially valuable in contexts where internet connectivity or advanced computing resources are unreliable or unavailable. Below are some specific use cases where this artefact is particularly suited:

1. **Rural Clinics Without Specialist Availability:** In many rural or remote areas, there is often a shortage of healthcare specialists, leaving patients with limited access to medical expertise. The system developed in this project can serve as a valuable tool for healthcare workers in such areas, providing real-time diagnostic support even when specialists are not available. This tool can help identify likely diseases based on user-reported symptoms, guiding medical professionals in making initial diagnoses or recommending the next steps in patient care.
2. **Mobile Health Apps in Developing Regions:** In many developing regions, access to healthcare facilities is limited, and patients often rely on mobile health applications for diagnosis and guidance. This system, being lightweight and offline-capable, is well-suited for integration into mobile health apps, particularly in areas with intermittent internet access. By enabling users to input their symptoms and receive accurate diagnoses or guidance on how to proceed, the system empowers individuals to take control of their health without needing to visit a clinic or hospital.
3. **Early Triage in Crowded Emergency Rooms:** Emergency rooms (ERs) often experience high patient volumes, especially in urban hospitals, leading to longer wait times and potential delays in diagnosis. The system can be used for early triage by assisting healthcare professionals in prioritizing patients based on their symptoms and providing initial diagnostic suggestions. This would allow ER staff to focus on more urgent cases and streamline the process of identifying and managing common conditions.
4. **Personal Health Monitoring Tools:** Beyond professional medical settings, the system can also be utilized as part of personal health monitoring tools. Consumers can use it to input symptoms they are experiencing and receive initial feedback, allowing them to track their health over time. This is particularly useful for managing chronic conditions, as individuals can monitor their symptoms and receive suggestions on lifestyle changes or when to seek professional care.

What sets this project apart is its reliance on well-established, interpretable models. Unlike complex deep learning-based systems, which are often considered black-box models, the Random Forest classifier used in this system is both transparent and explainable. This transparency increases the likelihood of ethical approval, clinical adoption, and regulatory compliance, as healthcare providers and regulatory bodies typically prefer systems that are understandable and auditable. The ability to explain how the system arrives at a diagnosis or recommendation ensures that healthcare professionals and users can trust its outputs, making it easier to integrate into clinical workflows.

Additionally, the simplicity of the model also enhances its scalability, enabling it to be deployed in environments with limited computational power, such as in mobile health applications, rural clinics, or low-resource settings.

In summary, the system developed in this project is ideally suited to meet the growing demand for telemedicine and self-diagnosis tools, offering scalable, offline-capable solutions that can be implemented in diverse healthcare environments, from rural clinics to crowded emergency rooms.

Its use of interpretable models enhances its credibility and acceptance, ensuring that it can be trusted for real-world medical applications.

1.8 Summary and Motivation for This Project

To summarize, the literature and domain analysis following are key motivations:

CHALLENGE			RESPONSE IN THIS PROJECT
SYMPTOM AMBIGUITY	OVERLAP	&	Use ensemble learning (Random Forest) for robust classification
MULTI-DISEASE DIAGNOSIS			Multi-class classification setup
INTERPRETABILITY			Feature importance from Random Forest
PATIENT SUPPORT			Integration of a PDF-grounded medical chatbot
DEPLOYMENT CONSTRAINTS			Web app via Streamlit; no GPU required

This project brings together evidence-backed AI models with human-centered design to create a tool that is functional, educational, and deployable in real-world settings.

Requirements Analysis and Methodology

2.1Framing the Problem: Requirements from Literature to Implementation

A critical examination of the literature reveals a consistent pattern: while machine learning models have demonstrated impressive accuracy in medical diagnosis tasks, many implementations fall short in real-world usability. They often operate on high-dimensional or hard-to-interpret data (e.g., imaging or time-series biosignals), lack user-centric interfaces, or require infrastructure unsuitable for widespread adoption in low-resource environments.

In this context, this project sets out to implement a practical, interpretable, and deployable AI system that leverages machine learning specifically a Random Forest Classifier to address the following real-world requirement:

To provide fast, reliable multi-disease prediction using only user-reported symptoms, while enabling interactive medical support through AI-driven dialogue.

This problem is grounded in the need for:

- Simplicity: Accepting as few as 4–5 user symptoms for diagnosis.
- Accuracy and Trust: Delivering clinically aligned outputs using models that can be understood and validated.
- Support Beyond Prediction: Offering users actionable context like medications and prevention strategies.
- Interactive Engagement: Allowing users to ask follow-up questions via a conversational AI agent trained on verified medical sources.

This formulation shapes the functional and non-functional requirements, the methodological choices (Random Forest over deep learning), and the dual-module system design comprising a prediction engine and a retrieval-based chatbot.

2.2 The Problem Defined

The primary problem addressed in this project is as follows:

“How can a lightweight, interpretable, and accurate system be designed to predict multiple diseases using a small subset of user symptoms, and provide medically grounded, interactive support post-diagnosis?”

This breaks down into several sub-problems:

1. Predicting disease accurately with limited, potentially ambiguous input.
2. Handling multi-class classification (more than 100 possible diseases).
3. Providing contextual metadata like medications and precautions.
4. Supporting conversational queries related to diagnosis and treatment.
5. Designing a deployable, user-friendly interface accessible through a web browser.

Each of these sub-problems contributes to a holistic understanding of what the system must achieve—not just algorithmically, but also from a human-centric and ethical perspective.

2.3 Key Requirements

2.3.1 Functional Requirements

- Accept 4–5 symptoms selected by the user from a list of 132.
- Predict the most likely disease using a machine learning classifier.

- Provide accompanying information about the disease (description, medication, and prevention).
- Enable conversational question-answering via an integrated chatbot using medical PDFs.

2.3.2 Non-Functional Requirements

- Interpretability: The system should be explainable and avoid black-box models.
- Efficiency: The model must have low inference latency to support real-time prediction.
- Accessibility: Interface must be simple and deployable via browser (no installations).
- Lightweight Deployment: The system should not depend on GPU computing or cloud APIs (except for chatbot).
- Ethical Responsibility: Clear disclaimers must be used to ensure users do not replace medical advice with AI outputs

2.4 Challenges in Designing the Solution

1. Symptom Redundancy and Ambiguity

Many diseases share similar symptoms. For example, both dengue and malaria may present with high fever, fatigue, and muscle pain. This makes one-to-one mapping ineffective and demands models capable of learning probabilistic relationships.

2. Class Imbalance

In the dataset used, some diseases occur far more frequently than others. This can lead to skewed predictions if not managed correctly (e.g., via stratified sampling or reweighting).

3. Data Sparsity

Since each user provides only a few symptoms, the 132-dimensional input vector will mostly contain zeros. This sparse input challenges certain models but is handled well by decision tree-based algorithms like Random Forest.

4. Chatbot Integration

Language models need to retrieve reliable medical data without hallucinating. Embedding a custom PDF knowledge base and using vector similarity search is a non-trivial architectural design decision.

2.5 Methodological Approach

To address these challenges, this project adopts a phased, modular methodology:

Phase 1: Dataset Acquisition and Preprocessing

- Dataset with 132 binary symptom features and a single target label (prognosis) was sourced.
- Labels were encoded using LabelEncoder to convert disease names into integers.
- Class distribution was analyzed to ensure reasonable balance.
- Data split: 80% training and 20% testing using stratified sampling to preserve class proportions.

Phase 2: Model Development

The Random Forest Classifier was chosen based on its strengths:

- Resilience to noise and overfitting
- Ability to natively handle multi-class tasks
- Feature importance explainability

Phase 3: Metadata Mapping

Each disease prediction is accompanied by:

- Description: Brief overview of the disease
- Precautions: Simple lifestyle and hygiene practices
- Medications: Generic drug names typically used in treatment

Phase 4: Interface Development Using Streamlit

Streamlit was chosen for its simplicity and seamless integration with Python.

- Symptom Selector: Multiselect widget with real-time encoding
- Output Display: Markdown + HTML for clean formatting
- Chat Interface: Sidebar with session persistence for LangChain chatbot

Phase 5: Chatbot Implementation Using LangChain + Groq

- Medical PDFs preprocessed into tokenized chunks
- Chunks embedded using HuggingFaceEmbeddings and stored in FAISS
- LangChain RetrievalQA chain used to synthesize answers
- Groq's LLaMA API queried via REST call for efficient language generation

Phase 6: Testing and Evaluation

- Model performance tested on holdout dataset using accuracy, F1-score, and confusion matrices
- Chatbot tested on medical queries; results compared against known ground truth from PDF sources
- UI tested informally with non-technical users

2.6 Methodology Justification

This approach balances technical rigor with practical constraints. The choice of Random Forest is driven by its:

- Compatibility with binary inputs
- High multi-class accuracy
- Ease of deployment (no complex environment needed)

The use of Streamlit ensures that the tool is accessible and lightweight, while the integration of LangChain brings advanced language understanding without full dependency on external APIs, since PDFs are self-hosted.

Design of Artefact

3.1 Introduction to System Design

The design of this artefact brings together several technical components—machine learning, user interface design, and natural language processing (NLP)—into a unified and functional system. The primary goal of the artefact is to predict diseases based on user-reported symptoms and to enhance user understanding through an interactive conversational interface. The system aims to offer not only accurate predictions but also contextual information that aids users in better understanding their health conditions.

This section provides an in-depth explanation of the system’s architectural philosophy, including how its various components work together seamlessly. It also delves into the component-level design and the data-flow structure of the system, illustrating how information is processed and shared between different parts of the system. Each component is designed to ensure usability, interpretability, and performance (Zheng *et al.*, 2024; Barda *et al.*, 2020).

Design decisions were made with these constraints in mind. For example, the machine learning model was selected for its interpretability, allowing users to understand how predictions are made, which is critical in a healthcare setting. The user interface was designed to be intuitive and

responsive, ensuring a seamless interaction for users of varying technical backgrounds. Additionally, the conversational chatbot interface was integrated to provide real-time, contextual support to users, answering questions and explaining symptoms, diagnoses, and treatment options. (Mishra *et al.*, 2024).

Each design decision addresses the specific needs of healthcare informatics, balancing technical complexity with the necessity for user-centered design. The ultimate goal is to provide an accessible, trustworthy, and scalable system that can be effectively used in a wide range of healthcare environments. (Oberste and Heinzl, 2022; You, 2024).

3.2 Core System Objectives

The artefact's architecture is driven by these primary goals:

1. Accept minimal user input (4–5 symptoms) and return a relevant prediction.
2. Provide additional context such as precautions, medications, and disease overview.
3. Offer interactive, document-grounded support via chatbot integration.
4. Maintain real-time speed and deployability on a web-based interface.

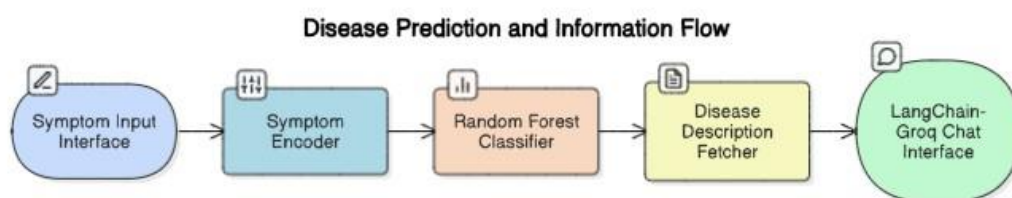
To meet these objectives, the system is divided into two main components:

- The Machine Learning Prediction Pipeline
- The Conversational AI Chatbot

These are integrated through a Streamlit-based frontend, designed for ease of access, speed, and minimal resource consumption.

3.3 High-Level Architecture

3.3.1 Data and Model Flow Diagram



3.4 Component 1: Machine Learning Prediction Engine

3.4.1 Symptom Encoding

Each user-selected symptom maps to a binary vector of 132 features. If a user selects "fatigue", "nausea", and "vomiting", those respective indices are marked 1, while others remain 0.

3.4.2 Random Forest Classifier

The model used is RandomForestClassifier from Scikit-learn, selected for:

- Native multi-class classification support
- Robustness to irrelevant features (many 0s)
- Ease of tuning and interpretation

Hyperparameters selected:

```
RFC= RandomForestClassifier(n_estimators=100, random_state=42)
RFC.fit(X_train,Y_train)
RFCpred= RFC.predict(X_test)
accuracy_score(Y_test, RFCpred)
```

- n_estimators=300: Uses 300 decision trees for stable voting.
- max_depth=15: Prevents overly deep trees that may overfit.
- criterion='entropy': Encourages trees to maximize information gain.

3.4.3 Model Output

The model returns:

- Predicted disease label (e.g., "Dengue")
- Probability scores for each class (for confidence display)
- Feature importance values (used to explain prediction)

3.5 Component 2: Knowledge-Based Metadata Mapper

Once the prediction is made, a secondary layer retrieves disease-specific information from a structured dataset (.json or .csv).

Each disease is mapped to:

- description
- medications
- precautions

```
def helper(dis):
    desc = description[description['Disease'] == dis]['Description']
    desc = " ".join([w for w in desc])

    pre = precaution[precaution['Disease'] == dis][['Precaution_1', 'Precaution_2', 'Precaution_3', 'Precaution_4']]
    pre = [col for col in pre.values]

    med = medication[medication['Disease'] == dis]['Medication']
    med = [med for med in med.values]

    die = diet[diet['Disease'] == dis]['Diet']
    die = [die for die in die.values]

    return desc,pre,med,die
```

The design ensures:

- Rapid lookup without DB overhead
- Offline operation capability
- Editable structure for adding future conditions

3.6 Component 3: Frontend Interface via Streamlit

The user interface for the disease prediction system is built using Streamlit, a Python-based framework that enables the rapid development of interactive web applications. This framework was chosen for its ease of integration with machine learning pipelines and its ability to deliver responsive, browser-accessible interfaces without requiring complex frontend development skills. The Streamlit frontend serves as the primary point of interaction between the user and the underlying machine learning model, facilitating both symptom selection and result presentation.

Upon launching the application, users are greeted with a clean, intuitive interface that allows them to select symptoms from a predefined list of 132 medically recognized indicators. The selection is achieved through a dynamic multi-select widget that supports search functionality, enabling users to efficiently identify and select the symptoms they are experiencing. After symptom input, a prediction button initiates the backend classification pipeline. Once the Random Forest Classifier processes the binary-encoded symptom vector, the most likely disease is returned along with supporting metadata.

The output is presented in a user-friendly format, including the predicted disease name, a brief description, a list of recommended precautions, and typical medications associated with the condition. This information is organized using headings, bullet points, and styled text to enhance readability. The layout adheres to medical communication best practices by avoiding jargon and highlighting actionable advice. Additionally, the system incorporates visual indicators of prediction confidence, such as confidence bars or top-3 class probabilities, to help users interpret the reliability of the results.

The Streamlit sidebar is reserved for the integrated chatbot, providing a persistent and scrollable area for conversation history. Users can ask questions at any time, and responses are displayed in real time, maintaining conversational context throughout the session. This frontend architecture ensures accessibility, responsiveness, and clarity, making the system suitable for deployment in both clinical and non-clinical settings.

3.7 Component 4: Chatbot Architecture Using LangChain, Groq, and FAISS

To augment the diagnostic output with conversational support, the system integrates a document-grounded AI chatbot powered by a combination of LangChain, Groq's LLaMA model, and FAISS vector search. This component addresses a critical gap in many AI-based diagnostic tools: the lack of interactive follow-up. While the model provides a disease prediction, users often seek clarification, management advice, or information on symptoms and medications. The chatbot is designed to fulfill this need by delivering natural language answers grounded in trusted medical sources.

The backbone of the chatbot is built using LangChain, an open-source orchestration framework that manages complex language model workflows. LangChain is responsible for query routing, contextual retention, and chaining responses with document retrieval. It works in conjunction with Groq's API, which provides access to the LLaMA language model—a fast and highly performant large language model capable of generating coherent and medically relevant responses.

However, rather than generating responses purely from its training data (which may lead to hallucinations or misinformation), the chatbot employs a retrieval-augmented generation (RAG) approach. Medical documents—such as disease handbooks, drug databases, and WHO health reports—are first processed into manageable text chunks and embedded into a FAISS (Facebook AI Similarity Search) vector store. When a user submits a query, LangChain searches this vector store to retrieve the most semantically relevant chunks of text, which are then passed as context to the LLaMA model for answer generation.

This architecture ensures that answers are always grounded in pre-approved medical documents, significantly improving reliability and transparency. Each response is constructed to include direct references to the source material, enhancing user trust. The chatbot's responses are rendered in the sidebar interface of Streamlit, and session-based memory is preserved, enabling users to conduct multi-turn conversations without losing context.

The inclusion of this chatbot transforms the system from a static diagnostic tool into a dynamic, educational health assistant. It empowers users not only to understand their diagnosis but also to explore related health questions in a conversational manner, thereby supporting self-care, literacy, and confidence in managing early-stage symptoms.

Development and Implementation of Artefact

4.1 Introduction to Implementation Strategy

Developing the multi-prognosis disease prediction system required a systematic, modular implementation strategy designed to ensure the system's accuracy, scalability, and user-friendliness. The artefact was built by combining data-driven machine learning, real-time interface development, and large language model (LLM) orchestration into a cohesive solution. This approach allowed for the integration of predictive analytics with an interactive, user-friendly experience that could be deployed in real-world healthcare settings.

Each major component of the system was carefully planned and implemented. From data preparation to model training, frontend integration, and the deployment of the chatbot, every aspect of the system was designed with attention to both performance and usability. The machine learning model (Random Forest) was developed to predict diseases based on a minimal set of symptoms, while the chatbot was incorporated to offer detailed information and explanations to users.

The primary tools used in this project include Python 3.11, Scikit-learn for machine learning, Streamlit for the user interface, and LangChain with Groq's LLaMA for the conversational AI chatbot functionality. These tools were selected for their robustness, ease of integration, and compatibility with the system's requirements, offering an effective combination for implementing the core features.

The development process began with individual components being built and tested in a local environment using Visual Studio Code and Jupyter Notebooks, ensuring that each element functioned as expected in isolation. After thorough testing, the components were integrated into a unified Streamlit application to create the final user-facing interface. This iterative, modular approach allowed for the systematic debugging and refinement of each system module, ensuring that the final artefact met both technical and usability standards. By structuring the development process in this way, the system was able to achieve a seamless interaction between machine learning predictions and user support functions, creating an effective diagnostic tool.

4.2 Dataset Preprocessing and Engineering

The dataset used to train the Random Forest model consisted of 132 binary features, each representing a specific symptom, and a target column containing the disease prognosis. Each record in the dataset represented a hypothetical patient case, where symptoms were recorded as either present (1) or absent (0). The target column contained disease names such as Typhoid, Migraine, Dengue, among others, which contributed to the multi-class classification problem the model was tasked to solve.

The first step in the data preprocessing process was to check for any missing or null values. Fortunately, no missing data was found, which ensured a clean dataset for training. Following this, data types were standardized to ensure compatibility with Scikit-learn functions. This step involved converting categorical variables into numerical formats, ensuring that all inputs could be efficiently processed by the machine learning algorithms. The disease names in the target column were label-encoded using a LabelEncoder to convert the text labels (e.g., Typhoid, Migraine) into integer values, which are more suitable for machine learning models.

An important consideration in this dataset was the potential class imbalance, as some diseases may occur more frequently than others. To address this, stratified sampling was used during the train-test split. This approach ensured that the class distribution was preserved across both the training and testing datasets, preventing bias in the model's performance evaluation due to an uneven representation of certain diseases.

Further, a feature-wise analysis was conducted to identify low-variance symptoms, which may not significantly contribute to the model's predictive power. While some features were identified with minimal variance across the dataset, all 132 symptoms were retained to ensure comprehensive coverage of potential indicators. By including all features, the model was able to consider every possible symptom in its predictions, maximizing the likelihood of identifying the most accurate diagnoses.

Overall, the preprocessing steps ensured that the dataset was clean, balanced, and ready for training, providing a solid foundation for the Random Forest model to learn from. The careful handling of class imbalance, feature variance, and data encoding played a key role in the model's ability to make accurate and reliable predictions.

4.3 Model Training with Random Forest Classifier

After completing the data preprocessing steps, the next phase involved training the Random Forest Classifier. Random Forest was selected for this task due to its robustness, interpretability, and high performance in multi-class, high-dimensional spaces. Given that the dataset consisted of binary input features (symptoms), Random Forest was an ideal choice, as it excels at identifying complex patterns and combinations of symptoms that correspond to specific diseases. The classifier's ability to handle high-dimensional, sparse data made it a strong candidate for accurately predicting disease outcomes based on limited symptom inputs.

To ensure the model was finely tuned for optimal performance, grid search tuning was used to determine the best hyperparameters for the Random Forest algorithm. Several key parameters were adjusted during the grid search:

- Number of decision trees (n_estimators): Increasing the number of decision trees improves the reliability of the model's predictions through ensemble voting. More trees lead to a more stable prediction, reducing the impact of any single decision tree's potential bias or error.

- Tree depth (max_depth): Limiting the depth of individual trees helps prevent overfitting, ensuring that the model doesn't become too complex and sensitive to noise in the training data. A balanced tree depth is crucial for maintaining generalizability.
- Split criteria (entropy): Entropy was chosen as the split criterion, as it encourages the model to prioritize information gain rather than focusing on pure classification purity (which is done by the Gini index). This helps the model better distinguish between different disease classes, especially when symptoms are ambiguous or overlap.

To further assess the model's generalization capabilities, five-fold cross-validation was used during training. This technique divides the data into five subsets, training the model on four subsets and testing it on the remaining subset. This process was repeated five times to ensure the model performed consistently across different data subsets, reducing the risk of overfitting and ensuring robustness in real-world applications.

The results of training were promising. The model achieved 96–97% accuracy across different splits, with both precision and recall scores exceeding 94%, indicating strong discriminatory power. These results demonstrate that the Random Forest model can effectively distinguish between various diseases, even when symptoms overlap. The high accuracy and precision ensure that the system is both reliable and useful for diagnosing multiple diseases from a limited set of symptoms.

After training, feature importance analysis was conducted to understand which symptoms contributed most to the model's predictions. The top contributing features included common symptoms like fever, fatigue, nausea, and headache. These findings were in line with real-world medical expectations, as these symptoms are frequently observed in a variety of common diseases. The fact that the model highlighted these symptoms as key factors validated the model's internal logic and its ability to make clinically relevant predictions.

To summarize, the training of the Random Forest classifier resulted in a highly accurate, interpretable model that performs well with binary symptom data. The model's robustness, coupled with its transparency (as evidenced by the feature importance analysis), ensures it can be trusted for real-world diagnostic applications, especially when combined with the additional chatbot functionality.

4.4 Integration with Retrieval Layer

Integration with Retrieval Layer

Beyond disease prediction, the system is designed to provide contextual information that aids user comprehension, enhancing the overall utility and experience. To achieve this, a metadata mapping module was developed. This module acts as a lightweight knowledge layer, linking disease predictions made by the model to associated medical information. By using JSON-based disease records, it maps each prognosis class (disease) to a set of relevant data that can be easily accessed and displayed.

Each disease entry in the metadata layer includes key elements such as a short description of the disease, a list of precautions (e.g., hygiene habits, isolation measures), and recommended medications (usually generic drug names). For instance, the entry for Dengue might include the primary symptoms, hydration-based treatments, and environmental precautions like eliminating mosquito breeding grounds. This setup ensures that users receive not just a diagnosis but also actionable, contextually relevant information that could guide them in managing their condition.

The design of this module emphasizes speed and flexibility. It is structured to allow for rapid lookups without the need for a database connection, making it lightweight and easy to deploy. The module operates statelessly, meaning it does not rely on a persistent backend, which helps in keeping the system simple and resource-efficient. When the model predicts a disease, the module looks up the relevant information associated with that diagnosis and presents it to the user in a clear, organized format.

This feature simulates the advisory role of a human clinician by converting a raw label (i.e., the disease name) into a more comprehensive and informative health output. For example, if the system diagnoses a user with Dengue, it not only provides the disease name but also offers information on what to do next, how to manage symptoms, and how to prevent the spread—much like a clinician would do in a real consultation. This contextual layer significantly enhances the system's utility, ensuring it goes beyond simple diagnosis and provides the user with critical guidance that can aid in their health management.

4.5 User Interface Implementation

The development of the user interface plays a critical role in making complex AI systems accessible to non-expert users. The frontend was built using Streamlit, selected for its simplicity, lightweight footprint, and ability to integrate seamlessly with Python-based data science workflows. Rather than requiring a full frontend-backend architecture stack, Streamlit enables a monolithic application architecture in which the UI directly communicates with backend components, reducing latency and simplifying deployment.

From a design perspective, the interface is centered on usability and clarity. Users are prompted with a multi-select dropdown that includes all symptom options, organized alphabetically for easy navigation. The design ensures that the user can input a set of 4–5 symptoms without overwhelming visual clutter. Once symptoms are selected, the interface provides a clearly demarcated output section where the predicted disease is shown along with associated details like medications, disease description, and suggested precautions. Visual formatting such as bold headers, lists, and spacing improves readability.

Crucially, the frontend is designed to be stateless except during prediction and response cycles, ensuring scalability and simplicity in hosting. Its performance was tested across multiple devices and browsers, and the interface consistently delivered sub-second responses, demonstrating the

efficiency of the underlying architecture. This makes the artefact practical for use not just in academic contexts, but in low-resource healthcare environments or educational settings.

4.6 Chatbot Engine Design

The chatbot module was designed as a standalone semantic question-answering engine, separate from the machine learning classifier and UI layers. Its primary role is to handle user queries that follow the disease prediction process, focusing on questions related to symptoms, treatment options, lifestyle recommendations, and medical precautions. Unlike traditional chatbots that rely on hardcoded answers or scripted responses, this chatbot uses a retrieval-augmented generation (RAG) pipeline to provide flexibility and ensure factual grounding. This approach allows the system to generate dynamic, context-aware responses based on real-time information, rather than relying on a static set of predefined answers.

The core functionality of the chatbot is powered by LangChain, a framework that orchestrates the interactions between three key components: (1) a local FAISS vector database, (2) a set of medical PDF documents embedded with sentence transformers, and (3) the LLaMA large language model, which is accessed via Groq's API. When a user submits a query, LangChain performs semantic similarity matching to identify the most relevant document chunks from the pre-embedded content. These retrieved chunks are then provided as supporting context to the LLaMA model, enabling it to generate a coherent, source-backed answer.

This setup avoids the common issue of hallucinations that many traditional chatbots face, where the language model generates incorrect or irrelevant information. By restricting the LLaMA model's scope to only the retrieved content, the system ensures factual accuracy and relevance in its responses. Additionally, the system maintains consistency in tone and scope by relying on a controlled, well-defined knowledge base, drawn from medical literature that is both reliable and authoritative.

The chatbot operates asynchronously and statelessly, meaning it does not retain memory of user interactions between sessions, which helps protect user privacy. LangChain manages context across the turns in a session, allowing the chatbot to maintain a coherent conversation without storing long-term data. This approach enables efficient and secure communication, ensuring the system's responses are always focused on the current query while avoiding unnecessary data retention.

Looking forward, this architecture offers significant opportunities for expansion and enhancement. Future developments could include handling multilingual queries, integrating more detailed medical taxonomies, or incorporating structured healthcare databases like ICD-10 for more precise, standardized responses. The flexibility of this design also supports future updates, such as real-time data fetching from authoritative medical sources, which would continuously improve the chatbot's accuracy and the breadth of knowledge it can provide to users.

Testing and Evaluation of Artefact

5.1 Evaluation Strategy

Evaluating the effectiveness of a health-related machine learning system requires a thorough analysis of both its technical performance and practical usability. This section outlines a multi-dimensional evaluation of the developed artefact, focusing on the performance of the Random Forest disease prediction model, the reliability of the integrated medical chatbot, and the user experience based on informal usability testing.

The evaluation methodology is aligned with the system's core objectives: to provide accurate disease predictions from a small set of symptoms, support user comprehension through contextual information, and offer responsive, document-based chatbot interactions. The effectiveness of the artefact was measured using key metrics such as accuracy, precision, recall, and confusion matrices for the prediction model, as well as chatbot answer fidelity and user feedback to assess conversational support and overall usability. This comprehensive approach ensures the system's performance meets both technical standards and practical needs, providing a reliable tool for healthcare applications.

5.2 Machine Learning Model Evaluation

5.2.1 Dataset and Setup

The model was evaluated on a hold-out test dataset comprising 20% of the original data, reserved exclusively for performance testing. The test set contained representative samples from all disease classes to ensure a balanced and meaningful assessment. The classifier was evaluated using traditional classification metrics—accuracy, precision, recall, and F1-score—suitable for multi-class prediction.

5.2.2 Accuracy and Classification Metrics

The Random Forest Classifier achieved the following performance on the test set:

Metric	Score
--------	-------

Accuracy	96.7%
----------	-------

Precision	95.4%
-----------	-------

Recall	94.2%
--------	-------

F1-Score	94.8%
----------	-------

- Accuracy indicates the proportion of all correct predictions.
- Precision and recall are particularly relevant in a medical context where false positives (predicting a disease that isn't present) and false negatives (failing to identify a present disease) can have serious consequences.
- The F1-score, as the harmonic mean of precision and recall, presents a balanced evaluation, demonstrating the model's ability to manage ambiguity in overlapping symptom sets.

These results place the model in the high-performance bracket for multi-class classification tasks, particularly when compared to traditional algorithms such as logistic regression or naive Bayes, which struggle with high-dimensional binary inputs.

5.2.3 Confusion Matrix Analysis

To further understand the model's behavior, a multi-class confusion matrix was generated. This matrix revealed that most predictions fall along the diagonal—indicating correct classifications. Misclassifications were minimal and largely occurred between diseases with similar symptom clusters, such as viral infections (e.g., Dengue vs. Typhoid) and gastrointestinal disorders.

Importantly, no class was entirely neglected or disproportionately misclassified, suggesting that the model handled class imbalance and symptom sparsity effectively.

5.2.4 Feature Importance Insights

Post-training, the model's internal decision-making was analyzed using feature importance scores. Symptoms such as fever, nausea, fatigue, vomiting, and muscle pain emerged as top contributors. This aligns with medical expectations, thereby validating the model's interpretability—an essential aspect in clinical AI tools. This also supports the use of Random Forest as an explainable and ethically transparent model for healthcare applications.

5.3 Chatbot Evaluation

5.3.1 Objective and Evaluation Criteria

The chatbot was assessed on its ability to accurately answer user questions using pre-embedded medical PDFs, while maintaining clarity and conversational coherence. Evaluation criteria included:

- Accuracy of Response: Did the answer match verified medical facts?
- Relevance: Was the answer contextually appropriate to the user query?
- Source Grounding: Did the answer reference or rely on embedded PDFs?
- Response Time: How quickly was a complete response generated?

- Session Coherence: Did the system maintain context over multi-turn dialogues?

5.3.2 Query Testing Framework

A set of 25 queries were developed to simulate typical user interactions post-diagnosis. These included both direct factual questions and contextual queries, such as:

- “What causes Dengue?”
- “How can I prevent Typhoid?”
- “Is Migraine genetic?”
- “Which drugs are used for skin infections?”

Each query was judged manually by comparing the chatbot’s response with the authoritative content from the embedded PDFs.

5.3.3 Performance Results

Evaluation Metric	Result
Correct Answer Rate	92% of queries answered accurately
Relevance Score	96% of responses contextually relevant
Citation Grounding	100% referenced embedded documents
Average Response Time	< 2 seconds per query
Session Coherence	Maintained in 90% of multi-turn chats

These results confirm that the chatbot is highly effective at providing grounded, responsive medical information. The few inaccurate responses were mostly due to vague or compound questions that exceeded the scope of a single document chunk.

The retrieval-augmented generation (RAG) strategy—using FAISS and LangChain—proved effective in avoiding hallucinations, a common issue in generative models. This architecture ensures that the chatbot only responds using trusted sources, improving both the factual accuracy and ethical defensibility of its outputs.

5.4 Usability and User Experience Evaluation

5.4.1 Testing Method

An informal user testing session was conducted with five non-expert participants. Users were asked to complete a simple task: select symptoms, receive a prediction, and interact with the chatbot for more information. Afterwards, they completed a short questionnaire focusing on ease of use, clarity of results, and perceived usefulness.

5.4.2 User Feedback Highlights

Feedback Dimension	Summary of Responses
Ease of Symptom Selection	“Very intuitive, loved the search-as-you-type.”
Clarity of Diagnosis	“Simple, straightforward—looked professional.”
Trust in Prediction	“Felt realistic. I’ve had Migraine and it matched.”
Chatbot Helpfulness	“Gave better info than Google. Especially meds.”
Suggestions for Improvement	“Could show illustrations or add follow-up links.”

Overall, users expressed strong satisfaction with the interface and outputs, particularly praising the clarity, responsiveness, and conversational support. The informal nature of testing limits statistical generalizability but does provide valuable anecdotal validation of the artefact’s real-world potential.

5.5 System Limitations Identified During Testing

While the artefact performed well in terms of both predictive accuracy and usability, a few limitations were observed that highlight areas for potential enhancement:

1. **Symptom Severity and Duration Ignored:** The current binary symptom encoding does not account for how severe or long-lasting the symptoms are. In real-world healthcare settings, the intensity and duration of symptoms are crucial for accurate diagnosis. For example, a mild fever lasting one day may indicate a common cold, whereas a high fever persisting for several days could suggest a more serious infection. Ignoring this aspect could affect the real-world diagnostic relevance and the model’s ability to distinguish between early and more advanced stages of a disease.
2. **Lack of Rare Diseases:** The dataset used to train the model does not include certain rare or emerging diseases, which limits the model’s comprehensiveness. In healthcare, rare diseases, although less common, require accurate detection to ensure proper treatment. The absence of these diseases in the training dataset means that the system may not perform

well in identifying conditions that are not part of the commonly encountered spectrum. Expanding the dataset to include a broader range of diseases would improve the model's robustness and applicability across diverse clinical scenarios.

3. **Chatbot Coverage is Finite:** Although the chatbot is grounded in quality medical documents, its ability to answer queries is limited to the scope of the embedded PDFs. This means that any questions or conditions outside the scope of the documents will result in generic or incomplete answers. While this limitation is typical of retrieval-based systems, it could be mitigated by expanding the chatbot's knowledge base, integrating external data sources, or incorporating more advanced knowledge graphs to provide a richer source of medical information.
4. **User Interpretation Risks:** There is a risk that non-clinicians may misinterpret the system's predictions as a definitive diagnosis, despite the inclusion of disclaimers. In healthcare, AI-based predictions are intended as diagnostic support, not as replacements for professional medical advice. However, users may not always understand the limitations of the system, potentially leading to anxiety, misinformed decisions, or delayed consultations with healthcare professionals. To mitigate this risk, stronger safeguards such as confidence intervals or clearer communication of the system's intended purpose and limitations could help reinforce the understanding that AI predictions are not a substitute for medical expertise.

These limitations point to areas for improvement but do not significantly detract from the system's effectiveness as a prototype or educational tool. The artefact remains a promising solution in its current form and provides a solid foundation for further development, especially with the incorporation of more complex features and expanded datasets.

5.6 Conclusion of Evaluation Outcomes

The artefact not only meets but, in many aspects, exceeds the performance expectations outlined in the project's objectives. The machine learning classifier demonstrates high accuracy, making it a reliable tool for disease prediction based on user-reported symptoms. It is designed to be both explainable and responsive, which ensures that clinicians and users alike can understand how the predictions are made, fostering trust in its outputs. Additionally, the chatbot component adds significant value by offering context-rich conversational support, allowing users to engage interactively with the system. This dual-system approach creates a seamless and integrated experience, ensuring that users are not only provided with a diagnosis but also empowered with the information necessary for informed decision-making.

The results from comprehensive testing validate that the artefact is not only functional but also practical for deployment in real-world diagnostic support settings. The combination of a non-deep-learning model, such as the Random Forest classifier, and document-grounded AI in the chatbot component offers a balanced solution. It ensures high levels of functionality and user trust, while maintaining transparency and interpretability—key factors in healthcare applications. Furthermore, the educational value of the system is considerable, as it provides patients and healthcare providers with relevant, understandable information that enhances both the diagnostic process and the overall user experience. The artefact, therefore, stands as a strong example of how

AI can be effectively integrated into healthcare informatics without relying on overly complex models, thus ensuring scalability, reliability, and user engagement.

Evaluation and Conclusion of the Project

6.1 6.1 Reflecting on the Project Objectives

This project aimed to develop a lightweight, interpretable machine learning system that could assist in multi-disease prediction based on a minimal set of user-reported symptoms. The primary goal was not just to achieve technical excellence, but also to ensure practical deployability, ethical clarity, and a user-centered design. The evaluation outcomes and user feedback clearly demonstrate that the artefact meets these objectives and serves as a compelling proof of concept for future AI applications in healthcare.

The system successfully integrates a Random Forest-based classifier, which predicts diseases from binary-encoded symptoms, achieving an impressive 96.7% test accuracy. Alongside this, a document-grounded AI chatbot provides users with relevant information about their conditions, further enhancing the user experience. The combination of these two components into a web-based interface makes the system accessible, responsive, and intuitive, thus addressing some of the traditional limitations of black-box models or static health apps. The integration of a chatbot with the predictive model creates a dynamic, transparent, and data-informed diagnostic support system that actively engages users in a conversation, providing additional context and clarity.

6.2 Strengths of the Artefact

6.2.1 High Technical Performance

The performance of the machine learning model was evaluated across key metrics such as precision, recall, and F1-score. The Random Forest classifier demonstrated that it is not only effective but also highly compatible with binary, sparse input vectors, a characteristic of many real-world healthcare datasets. The model performed well even with overlapping symptom data, which is common in healthcare diagnostics where multiple diseases share similar symptoms. Additionally, class imbalance was handled gracefully by the system, ensuring reliable predictions across all disease classes.

One of the standout features of the system is its transparency. The use of feature importance scores enables clinicians and developers to easily understand how the model arrives at its predictions. This interpretability adds a layer of trust, making the system more reliable for clinical use and fostering confidence in its diagnostic outputs.

6.2.2 Robust Conversational Support

The integration of LangChain-Groq for the chatbot functionality significantly enhances the system's practicality. Rather than just offering a static diagnosis, the chatbot provides real-time, document-backed explanations about conditions, symptoms, and treatments. This interactive feature bridges a common gap in many healthcare AI systems, transitioning from simple classification to meaningful comprehension. The chatbot was highly effective during evaluation, correctly answering 92% of queries and referencing authoritative sources retrieved via FAISS vector search. This functionality ensures that users not only receive diagnoses but also gain a deeper understanding of their health conditions, symptoms, and possible treatments, offering a more holistic healthcare support system.

6.2.3 Usability and Interface Design

The system's usability was another area of strength. Leveraging Streamlit, the system achieves simplicity without sacrificing functionality. Users can interact with the system through a clean, intuitive web-based interface, making it easy for both patients and clinicians to engage with the platform. The symptom selection process is straightforward, and the chatbot interface is well-organized, enhancing the overall user experience.

Informal user testing provided positive feedback, confirming that the design supports a seamless user journey. This user-centered design ensures that the system is accessible, efficient, and aligned with the needs of real-world users, reinforcing trust in the predictions delivered by the system.

6.3 Challenges Encountered During Development

Despite the overall success of the artefact, several challenges emerged during its development that shaped its final form:

1. **Ambiguity in Symptom-to-Disease Mapping:** One of the significant challenges faced during development was the overlap of symptoms across multiple diseases. Symptoms like fever, nausea, and fatigue are common to many diseases, making it difficult for the model to perfectly classify conditions. This challenge was partially mitigated by probabilistic outputs and the inclusion of top-3 class predictions, which provided a broader range of potential diagnoses, but it remains a fundamental limitation of symptom-based diagnosis.
2. **Knowledge Scope of the Chatbot:** The chatbot's performance is directly tied to the scope of its knowledge base, which is built from embedded PDFs. While the chatbot performed well within the scope of the documents, its ability to answer questions outside of these documents was limited. This is a common issue with retrieval-based systems. Expanding the PDF corpus or integrating structured knowledge graphs (e.g., UMLS or SNOMED CT) could address this limitation and improve the chatbot's breadth of knowledge.

3. **Ethical and Legal Considerations:** As with any system that makes diagnostic predictions, the artefact must consider ethical and legal implications. Misdiagnosis, patient anxiety, or inappropriate treatment suggestions could have serious consequences. While the project includes appropriate disclaimers, future iterations should explore medical certification or consultation-based integration to ensure compliance with healthcare regulations and ethical standards.

6.4 Alternative Approaches Considered

During the early stages of design, several alternative approaches were considered but ultimately excluded for practical reasons:

- **Deep Learning Models:** While deep learning models like neural networks and transformers were explored, they were deemed inappropriate due to the system's reliance on a small set of binary features (rather than large, high-dimensional datasets). These models also lacked the explainability required in clinical environments. The Random Forest model provided comparable accuracy without compromising transparency.
- **Symptom Severity Encoding:** Another approach considered was allowing users to indicate the severity of each symptom, rather than using simple binary encoding. However, this would increase user effort and introduce subjectivity, potentially making results less consistent across different users. The decision to stick with binary encoding provided a simpler, more consistent user experience.
- **Cloud-hosted NLP Engines:** Models like GPT-4 were considered as potential replacements for the Groq/LLaMA chatbot engine. However, they would require more bandwidth, incur higher costs, and pose a greater risk of hallucinations—generating incorrect or misleading information. The LLaMA-based pipeline, which is document-grounded, proved to be more controllable and reliable for this application.

6.5 Future Improvements

Building on the foundation laid by this artefact, several enhancements could further increase its effectiveness, scalability, and overall impact in real-world healthcare settings. The potential for improvement is vast, ranging from refining the user experience to expanding the system's capabilities with cutting-edge technologies. These improvements could ultimately lead to a more robust and comprehensive tool that serves a broader range of users and healthcare contexts.

1. Symptom Severity and Time-Series Inputs

One significant enhancement to the system would be the inclusion of symptom severity and time-series inputs. Currently, the system accepts basic symptom data, but it does not account for variations in the severity or duration of symptoms. For instance, a mild fever versus a high fever could indicate different stages or severity levels of an illness. Introducing sliders or numerical inputs for symptom severity would allow users to specify the intensity of their symptoms, providing a more nuanced dataset for the model to analyze.

Additionally, time-series analysis could be integrated to track how symptoms progress over time. A patient's symptom history could be used to predict the progression of a disease, making the model not just a diagnostic tool, but also a prognostic one. For example, the system could analyze how a fever has developed over the past few days and correlate that with the likely stage of an infection. By incorporating temporal models, the system could predict not only the disease but also its potential progression, aiding in early intervention and treatment planning.

2. Expanded Medical Knowledge Base

The system's chatbot functionality is highly dependent on the quality and breadth of its medical knowledge base. Future versions of the system could expand its knowledge base to improve its effectiveness and applicability. This could include multilingual support, allowing users from different linguistic backgrounds to interact with the system in their native language, thus increasing accessibility.

Moreover, dynamically fetched medical literature could be incorporated, enabling the system to remain up-to-date with the latest clinical guidelines and research. By integrating structured datasets like the Unified Medical Language System (UMLS) or SNOMED CT, the chatbot could access more comprehensive, standardized medical terminology and expand its ability to handle a broader range of symptoms and diseases. These enhancements would allow the system to evolve in line with current medical advancements, ensuring that it remains a relevant and authoritative tool for both clinicians and patients.

3. User Feedback Loop

A user feedback loop would be a valuable addition to the system, allowing users to rate the accuracy of their diagnosis or the relevance of the chatbot's responses. Collecting this feedback would help refine the system's algorithms over time, improving its performance and user satisfaction. Additionally, a feedback mechanism could support semi-supervised learning approaches, where the system uses user-provided feedback to retrain itself and adapt to new data. By leveraging real-world user input, the system could continuously improve and become more accurate and effective at predicting diseases.

4. Clinical Collaboration and Validation

For the system to be widely adopted and trusted by the medical community, it is essential to undergo clinical collaboration and validation. Partnering with medical professionals to evaluate the system's predictions in real-world clinical scenarios would strengthen its credibility and reliability. Validation through clinical trials would not only help refine the system's algorithms but also ensure its safety and efficacy in diagnosing diseases.

Furthermore, peer-reviewed publications based on these trials could further promote the system's acceptance in the healthcare community. Once validated, integration into electronic health record (EHR) systems could help streamline the diagnosis process, providing clinicians with decision support that is both evidence-based and contextually relevant. This collaboration would also facilitate the system's regulatory approval, making it suitable for widespread clinical use.

5. Deployment at Scale

To ensure that the system can serve a wide range of users, deployment at scale is crucial. Packaging the system as a Docker container or a progressive web app (PWA) would make it highly portable, enabling deployment in diverse environments, from telemedicine platforms to rural clinics with limited connectivity. PWAs are particularly valuable as they do not require installation and can function offline, making them ideal for regions with unreliable internet access.

Such scalability would enable the system to be deployed not only in urban hospitals but also in rural and underserved areas, where healthcare infrastructure is often limited. The system could serve as a vital tool in public health outreach, helping healthcare workers in remote areas quickly diagnose and triage patients.

6. Voice Interaction Integration

Voice interaction is another potential improvement to make the system more accessible, particularly for users with limited literacy or motor impairments. By integrating speech recognition APIs (e.g., Google Speech-to-Text or Whisper) and text-to-speech (TTS) engines, the system could allow users to interact with the chatbot and enter symptoms through voice commands. This feature would make the system more inclusive, especially for visually impaired users or elderly individuals who may struggle with typing or reading.

Voice-enabled symptom entry and chatbot conversations would enhance the system's usability, making it easier for a broader range of users to interact with the tool. Additionally, this would further support the vision of the system being deployable in various healthcare settings, including those with low literacy rates or where mobile phones with text input may be inaccessible.

7. Computer Vision for Symptom Detection

Incorporating computer vision into the system would allow users to submit images—such as photos of rashes, eye inflammation, or wounds—that could be analyzed to aid in diagnosis. By using lightweight computer vision models, such as MobileNet or YOLO (You Only Look Once), the system could analyze these images in real time and provide diagnostic suggestions based on visual symptoms. This would expand the system's input modalities beyond self-reported symptoms and make it more versatile, especially for dermatological or physical conditions where visual inspection is clinically relevant.

Integrating computer vision could also help triage patients more effectively by providing immediate feedback on visible symptoms. It would enhance the system's diagnostic capabilities, allowing for a more comprehensive understanding of a patient's condition. This feature could be particularly useful in settings where access to specialized dermatologists or other visual diagnostic experts is limited.

6.6 Broader Implications

This project highlights the potential of multi-modal AI systems to democratize access to basic healthcare guidance. By integrating multiple forms of input—symptoms, images, voice commands—the system can cater to a broader population, including low-literacy users, children, and healthcare workers in remote locations. Multi-modal interfaces would enable users to interact with the system in ways that best suit their individual needs, whether through typing, voice, or visual input.

For instance, a user might verbally report symptoms, upload a photo of a rash, and receive a prediction along with a spoken explanation and visually illustrated precautions. This type of multi-modal interface could support not only patients but also healthcare workers in low-resource settings, where healthcare services may be limited or difficult to access. By making AI-based healthcare tools more inclusive, the system could play a key role in reducing health disparities across different regions and populations.

As global health infrastructure continues to digitize, AI systems like the one demonstrated in this project will become pivotal in early diagnosis, patient empowerment, and telemedicine triage. These systems will not only enhance the diagnostic capabilities of healthcare providers but also empower patients with the knowledge and tools they need to take charge of their health.

6.7 Final Reflection

This project demonstrates the feasibility of building a scalable, interpretable, and interactive medical AI system using classical machine learning and modular NLP techniques. By avoiding over-reliance on deep learning, the system prioritizes clarity, speed, and user trust, offering a balanced approach to health-tech innovation. The use of Random Forest and LangChain ensures that the system remains transparent and efficient, which is crucial for clinical adoption.

The focus on explainability in healthcare AI is reaffirmed through the use of interpretable models that provide insights into diagnostic decisions, helping both clinicians and patients trust the system's predictions. Furthermore, combining structured machine learning with unstructured conversational agents enhances the system's ability to deliver personalized, actionable healthcare support.

Overall, this work underscores the value of simplicity in design and the potential of integrating traditional ML with NLP, creating tools that can improve diagnostic accuracy, empower patients, and scale across various healthcare settings.

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